

Name of the subject: **Surveying and Measured Drawing**

Department: **Architecture and Regional Planning**

Curriculum: **UG (2nd Year)**

Semester: **3rd (AUTUMN),**

LTP: **0-0-3,**

Credits: **2**

Type of Course: **Compulsory**

Prerequisite(s): **Nil**

Course Overview:

The course is mainly oriented to understand the basic surveying principles and measured drawing. The course is broadly subdivided into two major parts. Initially the basic concept of land surveying will be explained through Electronic distance measurement (EDM) instruments. The student will learn advanced surveying terms using a total station. The concept will be discussed through lecture sessions and followed by, hands-on field exercises to help acquire the practical understanding and basic skills needed to work in the surveying industry. In the second part of the course, the emphases will be given in the hands on training regarding the measured drawing exercise. The measured drawing of any heritage / vernacular building and the survey of existing settlement will be executed as assignment. The learning of basic principles and strategies of measured drawing will help the student to document the historical buildings in future.

Learning Objectives:

After the completion of this course, students will be able to:

1. Apply the principle of surveying techniques for distance and bearing measurements
2. Conduct the fly and profile levelling on ground.
3. Interpret the existing site data and convert into the contour map.
4. Perform the area and volume measurement from the survey sheet.
5. Establish the plan and elevation of existing building through measured drawing exercise.
6. Perform the field measurement and photographic survey to produce a settlement survey

Course Contents:

Principles of surveying, Methods of determining angles and directions. Distance measurement: Tacheometric Measurement and Electronic distance measurement (EDM), Total Station Survey and its multi-functioning in surveying Principle of levelling: Differential and Reciprocal levelling, Applications of levelling (profile levelling and contouring). Computation of Areas and Volumes: Area between traverse and irregular boundary, Areas and Volumes of Cross Section.

Measured drawings of selected heritage building / vernacular building/ settlement, utilising various survey tools techniques, Field measurement and data interpretation, Photo Survey.

List of Experiments (weekly)

Sl No	Topic
1	Taking horizontal distance and offsets (Perpendicular and Oblique) in different field conditions.
2	Determine included angles and measurement of bearings and closing error.
3	Carry out fly levelling in different field conditions.
4	Carry out profile levelling in different field conditions
5	Carry out the contour survey on an undulated ground and prepare the contour sheet
6	Worksheet on computation of Areas and Volumes
7	Introduction to Electronic Distance Measurement: height and distance Measurement, Survey Data Post Processing, Survey Data Applications.
8	Electronic Distance Measurement Survey, Area Measurement, sit out development
9	Introduction to Architectural Measured Drawing: development of a simple building plan
10	Measured drawing: Development of plan of a heritage / vernacular building
11	Measured drawing: Development of elevation and details
12	Field measurement and photographic survey: Development of a small settlement site plan

Text Book

1. T. P. Kanetkar and S.V.Kulkarni, *Surveying and Levelling Vol. I and Vol. II*, Pune Vidyarthi Griha Prakashan.
2. Subramanian, *Surveying and Levelling*, Oxford University Press.
3. B. C. Punmia, Ashok K. Jain, Arun K. Jain, *Surveying, Vol. I & II*, Laxmi Publications.

Reference books:

1. C. Venkatramaiah, *Textbook of Surveying*, University Press
2. John Uren & Bill Price, *Surveying for Engineers*, Palgrave Macmillan.
3. S. K. Duggal, *Surveying, Vol. I & II*, TataMc-Graw Hill.